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Chapter 1

INTRODUCTION

A general deterioration in memory and other cognitive abilities that is severe enough to impair an individual's capacity to carry out daily tasks is referred to as dementia. The steady and continuous decline in cognitive function is its defining feature. Individuals suffering with dementia experience difficulties with their thinking, actions, and day-to-day functioning. Patients with this condition also have memory loss and lack of understanding of their concerns. Machine learning (ML) and deep learning (DL) have become formidable instruments in the healthcare industry in recent times, transforming the methods of disease diagnosis, patient outcome prediction, and customized treatment regimens. Massive volumes of medical data, such as genomic sequences, wearable sensor data, electronic health records (EHRs), and medical photographs, can be analyzed by ML and DL algorithms to find patterns, extract insights, and make predictions that were previously difficult or impossible for human experts. A common and crippling neurological disorder called dementia is marked by a deterioration in cognitive ability that is severe enough to interfere with day-to-day activities. The most prevalent type of dementia, Alzheimer's disease, affects millions of people globally. Dementia is a growing burden on healthcare systems and communities as the world's population ages, underscoring the critical need for efficient diagnosis methods and interventions.

1.1 Machine Learning

A subfield of Artificial Intelligence called Machine learning (ML) gives computer systems the ability to learn from data and gradually get better at what they do without the need for explicit programming. By enabling algorithms to recognize patterns, anticipate outcomes, and extract insights from intricate information, it is revolutionizing various industries and altering the way humans engage with technology.

Machine learning, at its foundation, uses statistical methods to examine data and derive relevant knowledge. Machine learning algorithms have the ability to identify patterns and relationships in vast amounts of data that may not be visible to human observers. Through continual refinement of their predictions and decision-making skills, these algorithms learn from their experiences and adjust accordingly.

Different activities and learning objectives require different kinds of machine learning algorithms. Algorithms are trained on labeled data in supervised learning, where each sample is matched with the appropriate result. On the other hand, unsupervised learning entails using unlabeled data to learn about hidden structures or patterns. By fusing aspects of supervised and unsupervised learning paradigms, semi-supervised learning and reinforcement learning provide even more versatility.

1.2 Deep Learning

A form of machine learning called "deep learning" seeks to process data and identify patterns by simulating the neural networks seen in the human brain. Speech recognition, natural language processing, and computer vision are just a few of the fields it has transformed.

Artificial neural networks, which are made up of linked layers of nodes, or neurons, arranged into input, hidden, and output layers, are the fundamental building blocks of deep learning. Every neuron takes in information, processes it, and then sends its result to the layer above. These networks continually improve their performance over time by learning to modify the connections between neurons based on the mistake in their predictions through a process known as backpropagation.

Deep learning's capacity to automatically learn hierarchical data representations is one of its main features. Put more simply, it can recognize intricate features from unprocessed data without requiring intentional feature engineering on the part of humans. Deep learning models, for instance, may automatically pick up on edges, textures, and more complex notions like objects or scenes in picture recognition tasks.

1.3 Applications of Machine Learning & Deep Learning

One of the most fascinating technologies that people have ever encountered is machine learning. It provides the computer the ability to learn, which is apparent from the name and more resembles human intelligence. There are probably a lot more applications for machine learning than one might think.

1.3.1 Image Recognition

In image identification tasks, machine learning and deep learning techniques are frequently used, allowing computers to recognize and categorize objects inside digital photographs. Because convolutional neural networks (CNNs) can automatically learn hierarchical representations of visual information from raw pixel data, they are very effective in this domain. Medical image analysis, object detection, and facial recognition are a few examples of applications.

1.3.2 Fraud Detection

Have you ever had a transaction flagged by your bank as suspicious before you even knew something was off? That's because sophisticated algorithms that detect fraudulent activities in the background are to blame. Like a watchful security guard, they examine your purchasing patterns and transaction history to safeguard you against identity theft and credit card fraud.

1.3.3 Pattern Finding

Just picture searching through reams of data for a needle in a haystack—that is, a significant insight concealed there. We can find patterns and trends in all that data that could have gone missed without the aid of machine learning. It's similar to having an extremely skilled investigator who can find patterns and hints in even the most complicated datasets.

1.3.4 Disease Detection

Consider machine learning as a kind of virtual physician that can examine test data, patient histories, and medical scans to identify early warning indicators of disease. These algorithms assist medical personnel in taking early action, thereby enhancing patient outcomes and maybe saving lives, by detecting anomalies and warning indicators.

1.3.5 Automobile Anomaly Detection

Machine learning algorithms can identify irregularities in automotive systems and components, just like you would notice if your automobile started making unusual noises. They can assist keep your automobile operating smoothly and securely by identifying possible problems before they become serious ones by evaluating sensor data and diagnostic codes.

1.3.6 Weather Forecast

Ever ponder how weather forecasters make their predictions? They examine data from atmospheric models, weather stations, and satellites using complex algorithms. Whether it's rain, snow, or sunshine, these algorithms alert us to the weather by analyzing data and looking for patterns. It's similar to having a reliable weather forecaster on your side during good times and bad.

1.4 Project Introduction

Dementia is a complicated brain disorder that affects a lot of people all around the world. In the field of medicine, early detection is still a crucial yet challenging task. The primary focus of this initiative is the pressing need to enhance dementia early detection. With the use of technology developments, especially in MRI imaging and data analytics, the initiative seeks to develop a predictive model that is more accurate. The project focuses on combining two essential elements: the examination of patient-recorded data using classification algorithms and the use of convolutional neural networks (CNN) to analyze MRI scans in order to identify possible brain-related signs. Through the integration of several disparate but complementary approaches into a single platform, the project aims to improve dementia onset prediction accuracy and enable preventive healthcare interventions.

Early dementia detection is extremely important for patients and healthcare systems alike. By providing a more sophisticated predictive model, the combination of patient-specific data and state-of-the-art MRI insights seeks to address this issue. This methodology not only exhibits potential for enhanced predictive precision but also has the ability to optimize the diagnostic procedure, hence facilitating timely interventions and customized care regimens. The potential to transform early dementia detection and improve patient outcomes is highlighted by the thorough integration of modern imaging and patient data into a single platform.

1.5 Problem Statement

Early detection of dementia presents a formidable challenge in healthcare, necessitating innovative approaches for improved identification.

This project specifically targets the intricate task of early dementia detection by integrating two distinct predictive models. The first model delves into the intricate analysis of MRI scans using advanced convolutional neural networks (CNN). Meanwhile, the second model focuses on mining patient-recorded data, employing classification algorithms like Naive Bayes or Random Forests. Both models operate within a cohesive system, aiming to enhance the precision and timeliness of dementia prediction.

Chapter 2

LITERATURE SURVEY

A literature survey or a literature review in a project report shows the various analyses and research made in the field of interest and the results already published, taking into account the various parameters of the project and the extent of the project. Literature survey is mainly carried out in order to analyze the background of the current project which helps to find out flaws in the existing system & guides on which unsolved problems we can work out. So, the following topics not only illustrate the background of the project but also uncover the problems and flaws which motivated to propose solutions and work on this project.

A literature review is a text of an academic publication that summarizes the state of the art on a given subject, encompassing theoretical and methodological contributions as well as substantive discoveries. Reviews of the literature report on fresh or original experimental work, but they do so via secondary sources. Although it's not always the case, a literature review is typically found before the methodology and results portion of academic literature, such as thesis, dissertations, and peer-reviewed journal articles. Reviews of previous research are also frequently seen in search proposals and prospectuses, which are approved documents that are submitted before a student starts a dissertation or thesis. Its primary objectives are to give the reader with context and to place the current study within the larger body of literature. Literature reviews are a basis for researching nearly every academic field. A literature survey includes the following:

- Recognized theories that are currently in existence regarding the subject.
- Books, both general and specialized, on the subject.
- Field research is often conducted in chronological order.
- The difficulties encountered and ongoing work, if any.

Objectives of Literature Survey

- Acquiring knowledge of the idea definitions.
- Access to the newest ideas, theories, and techniques.
- Discovering research topics based on the existing research
- Focus on your area of expertise. Words used in one field may not always mean the same thing in another.
- Eliminating distractions enhances the quality of the literature review. Don't forget to explain

what is not included.

[1] The research conducted by Varun Jain et al. in 2021 presents a novel **“AI-based system for the detection and severity prediction of dementia using MRI scans”**. The study introduces a DCGAN-based Augmentation and Classification (D-BAC) model approach to identify and classify dementia into various categories based on its severity in MRI scans. The investigation also includes the detection of Mild Cognitive Impairment (MCI) using a GAN-augmented dataset. The proposed model demonstrates a 74% accuracy in predicting MCI and can classify dementia into four categories based on MRI scan features. Visual Explainable AI (XAI) techniques, such as GradCAM, are utilized to interpret the model's internal workings and identify differentiating features in MRI scans learned during training.

The methodology employed in this study involves training a Convolutional Neural Network (CNN) model on three datasets: the original dataset, geometrically transformed images, and the GAN-augmented dataset. Performance analysis reveals the superiority of using the GAN-augmented dataset, showcasing the model's enhanced capabilities in dementia classification. The results indicate promising accuracy rates in predicting MCI and classifying dementia severity levels, highlighting the potential of AI in early dementia detection and severity assessment.

The study by Jain et al. in 2021 addresses the critical problem of imbalanced datasets in dementia classification by utilizing GAN augmentation to balance class labels and create a newly balanced dataset. The GAN-augmented dataset outperforms geometric transformations, as it maintains spatial conformity in images, aiding the CNN model in generalizing well to unseen test samples. The accuracy achieved using the GAN-augmented dataset is reported as 81% with a conventional CNN model and 86% with VGG-19, representing a significant improvement over existing state-of-the-art models that achieved only 49% and 52% accuracy. The model's ability to predict MCI with 74% accuracy further underscores its potential in early dementia detection.

Despite the promising results, the study acknowledges certain limitations. These include the necessity for a balanced dataset and a sufficient number of MRI images, which may not always be readily available in the medical domain. The performance of the model could be impacted by the lack of augmentation techniques to address class imbalance issues commonly found in medical datasets. Future research directions could focus on overcoming these limitations to further enhance the accuracy and applicability of the proposed AI-based system for dementia detection and severity prediction.

[2] Ho et al. (2023) conducted a study focusing on developing **“An app for predicting patient**

dementia classes using convolutional neural networks (CNN) and artificial neural networks (ANN)". The research methodology involved assessing outpatients with the clinical dementia rating instrument, collecting demographic data anonymously, and analyzing the data using contingency tables to establish relationships between CDR scores and frequency in the sample. The study utilized a chord diagram and radar plot for visualizing the correlation between CDR scores and patient assessments. The ANN algorithm was employed for model building, incorporating neuron stems and sigmoid functions to transform probabilities into values between 0 and 1.0.

The abstract highlights the study's objective of enhancing prediction accuracy through the app's development and evaluation. The app demonstrated a high prediction accuracy of 93.72% in classifying dementia classes among patients. The CNN and ANN models showed promising results in predicting dementia based on the clinical data collected. The radar plot visualization technique contributed significantly to improving prediction accuracy, offering a novel approach for dementia classification in patients responding to the CDR instrument.

Despite the promising results, the study identified several limitations that need to be addressed in future research. The comparison was limited to CNN and ANN models, suggesting the exploration of additional algorithm models for dementia prediction. The sample size used for the ANN model was relatively small, indicating the necessity for larger datasets to validate the findings. Furthermore, the study highlighted challenges in estimating a higher number of parameters using MS Excel's Solver add-in function, emphasizing the need for advanced tools for complex analyses.

In conclusion, Ho et al. (2023) presented a comprehensive study on predicting patient dementia classes using CNN and ANN models integrated with radar plot visualization. The app's high accuracy in classifying dementia classes signifies its potential for personalized treatment strategies. However, addressing the study's limitations and exploring diverse algorithm models for dementia prediction are crucial for advancing the field and improving prediction accuracy in clinical settings.

[3] The article "**Dementia Prediction Using Machine Learning**" by Sara Dhakal et al. published in *Procedia Computer Science* in 2023 presents a study on predicting dementia using machine learning techniques. The research aims to develop a model for dementia prediction based on features such as 'CDR', 'MMSE', 'eTIV', 'Gender', 'EDUC', and 'SES' selected through the chi-square method. The methodology involved data collection from the OASIS project, which provided MRI data of 416 subjects across different age groups. The dataset was pre-processed

through data visualization, imputation, and transformation stages before model building.

The abstract highlights the use of machine learning algorithms, including AdaBoost, Decision Tree, Extra Tree, Gradient Boost, K-NN, Logistic Regression, Naïve Bayes, and Random Forest, for dementia prediction. The study focuses on evaluating the accuracy, precision, F1-score, recall, and false positive rate of these algorithms. The results indicate that K-NN achieved the highest accuracy of 95.16%, followed by Logistic Regression and Naïve Bayes with an accuracy of 94.35%. These findings demonstrate the effectiveness of machine learning in predicting dementia based on the selected features.

However, the study has certain limitations that should be considered. One limitation is the reliance on a specific dataset from the OASIS project, which may not fully represent the diversity of dementia cases in the general population. Additionally, the study's focus on a limited set of features may overlook other important variables that could enhance the prediction accuracy of the model. Furthermore, the generalizability of the findings may be limited to the specific context and characteristics of the dataset used in the study.

In conclusion, the research by Sara Dhakal et al. provides valuable insights into the application of machine learning for dementia prediction. By utilizing features selected through the chi-square method and evaluating various machine learning algorithms, the study demonstrates promising results in terms of accuracy. Despite the limitations, the findings contribute to the growing body of research on using machine learning in healthcare for predictive analytics. This study serves as a foundation for further exploration and refinement of machine learning models for dementia prediction, with potential implications for early diagnosis and intervention in clinical practice.

[4] In the study "**Predem: A Computational Framework for Prediction of Early Dementia**" by Deep Sharma et al., researchers present a novel computational framework aimed at predicting early dementia using deep neural networks. The methodology employed in this research involves the utilization of various datasets, pre-processing techniques, and machine learning models to extract essential characteristics that can differentiate between individuals with dementia and healthy subjects. Specifically, deep learning models such as VGG-16, CNN, and LSTM-inspired RNN are utilized to forecast different phases of Alzheimer's disorder, potentially aiding in the prevention of dementia. The authors emphasize the significance of these models in leveraging image datasets without the need for manual feature extraction, making them particularly suitable for object detection and classification tasks.

The abstract of the study highlights the importance of early detection and prediction of dementia, a pressing issue in healthcare and cognitive research. By leveraging advanced computational

frameworks and deep neural networks, the researchers aim to enhance the accuracy and efficiency of diagnosing dementia at its early stages, thereby enabling timely interventions and treatments. The study underscores the potential of deep learning models in improving the predictive capabilities for identifying individuals at risk of developing dementia, ultimately contributing to the advancement of healthcare practices and patient outcomes.

In terms of results, the research demonstrates promising accuracy rates in predicting early dementia using the Predem computational framework. By splitting the training and testing datasets in an 80:20 ratio, with 80% for training and 20% for testing, the deep learning models exhibit notable performance in forecasting different phases of Alzheimer's disorder. The utilization of CNN, VGG-16, and LSTM-inspired RNN models showcases the effectiveness of these approaches in distinguishing between individuals with dementia and healthy subjects, thereby offering valuable insights for early detection and intervention.

However, despite the advancements and successes achieved in this study, certain limitations should be acknowledged. These may include constraints related to the size and diversity of the datasets used, potential biases in data collection, and the generalizability of the computational framework across different populations and settings. Addressing these limitations and conducting further validation studies will be crucial in enhancing the robustness and applicability of the Predem framework in real-world clinical settings.

In conclusion, the Predem computational framework developed by Deep Sharma et al. represents a significant contribution to the field of dementia prediction and early detection. By leveraging deep neural networks and advanced machine learning techniques, the study offers a promising approach to improving the accuracy and efficiency of diagnosing dementia at its early stages, with implications for enhancing patient care and outcomes in healthcare settings.

[5] In the systematic review titled "**Machine Learning for Dementia Prediction: A Systematic Review**" by Anderber et al. (2023), the authors investigate the application of machine learning (ML) and deep learning (DL) algorithms in predicting dementia and its subtypes using clinical data modalities such as images, clinical variables, and voice data. The study aims to analyze the performance of ML models in identifying variables that could aid in dementia prediction and disease progression forecasting for patients with mild cognitive impairment (MCI). Through a comprehensive examination of 75 research articles published in the last decade, the researchers highlight the increasing interest in developing automated diagnostic systems based on ML and DL techniques within the healthcare industry.

The methodology employed in the review involved the selection of research articles using

keywords related to ML, DL, and dementia subtypes (Alzheimer's disease, vascular dementia, frontotemporal dementia, and mixed dementia). The authors conducted an electronic search across various databases, resulting in the identification of 450 studies. After screening and eligibility assessments, a final set of 75 research papers were included in the analysis, with each data modality (image, clinical variables, voice) comprising 25 papers each. The study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a rigorous and systematic approach to article selection and analysis.

The results of the systematic review revealed that ML models exhibited high accuracy in distinguishing dementia patients from healthy individuals and in predicting disease progression for MCI patients. The authors noted a significant performance improvement in ML algorithms for dementia prediction over the past four years, as evidenced by an exponential increase in research publications utilizing ML techniques. However, the review emphasized the importance of interpreting accuracy metrics cautiously, as the prevalence of the illness in the target population can influence the overall accuracy of a diagnostic model. While ML models with higher sensitivity and specificity generally lead to increased accuracy in balanced datasets, the accuracy of a test for a rare ailment may be compromised despite high sensitivity and specificity.

Despite the promising results, the study identified several limitations in the existing research on ML-based dementia prediction. These limitations include the need for further exploration of feature extraction and selection methods, the adoption of diverse performance evaluation measures, and the assessment of ML model performance across various data types. Additionally, the review highlighted the necessity of addressing weaknesses in previously proposed ML models for dementia prediction to enhance the reliability and generalizability of automated diagnostic systems in clinical practice

Chapter 3

DESIGN AND METHODOLOGY

3.1 Design

The system “design” is defined as the process of applying various requirements and permits its physical realization. Various design features are followed to develop the system the design specification describes the features of the system, the component or elements of the system and their appearance to the end-users. The system design process comprises several sequential stages aimed at translating diverse requirements and objectives into a tangible plan for developing a functional system. Initially, in the analysis phase, stakeholders' needs, including functional and non-functional requirements, are meticulously gathered and assessed. Subsequently, during conceptualization, designers engage in creative exploration, brainstorming various ideas and alternatives to effectively address the identified requirements. Following this, in the detailed design phase, high-level concepts are refined into precise specifications, determining the system's architecture, components, interfaces, and functionality.

3.2 Design Constraints

There are requirements that must be met for a project to succeed, including design restrictions. When developing a project, design limitations aid in streamlining decisions. Many variables, including budgetary restrictions, time limits, technology limitations, legal obligations, environmental concerns, and resource availability, can lead to constraints. Though they can seem like a bad thing, design limitations really assist the project be tailored to the client's particular requirements. For engineers, designers, and architects to provide solutions that are workable, efficient, and practical within the limits set forth, they must comprehend and be able to operate within design constraints. Limitations frequently spur innovation because they force innovative problem-solving and the optimization of design decisions.

- Safety: In our project we are implementing with the tensorflow environment so it provides high security.
- Cost: In our project we are using normal working laptop with mid-range processor so cost also is low.

- **Quality:** In our project we are using advanced algorithms to detect so the quality also very high.

3.3 System Design

Our dementia detection system walks administrators through the procedure with ease, much like a trustworthy assistant. They input MRI pictures and patient data via a straightforward interface, and sophisticated models carefully analyze the data. Like devoted professionals, these models operate independently before combining their results to provide a definitive diagnosis. The system indicates more testing if there are any doubts, demonstrating its careful attention of accuracy. Transparency and cooperation are ensured by respecting patient privacy at all times and creating reports that are simple to share with medical experts. The goal of this strategy is to protect patient privacy while delivering trustworthy, reliable data.

3.3.1 Use Case Diagram

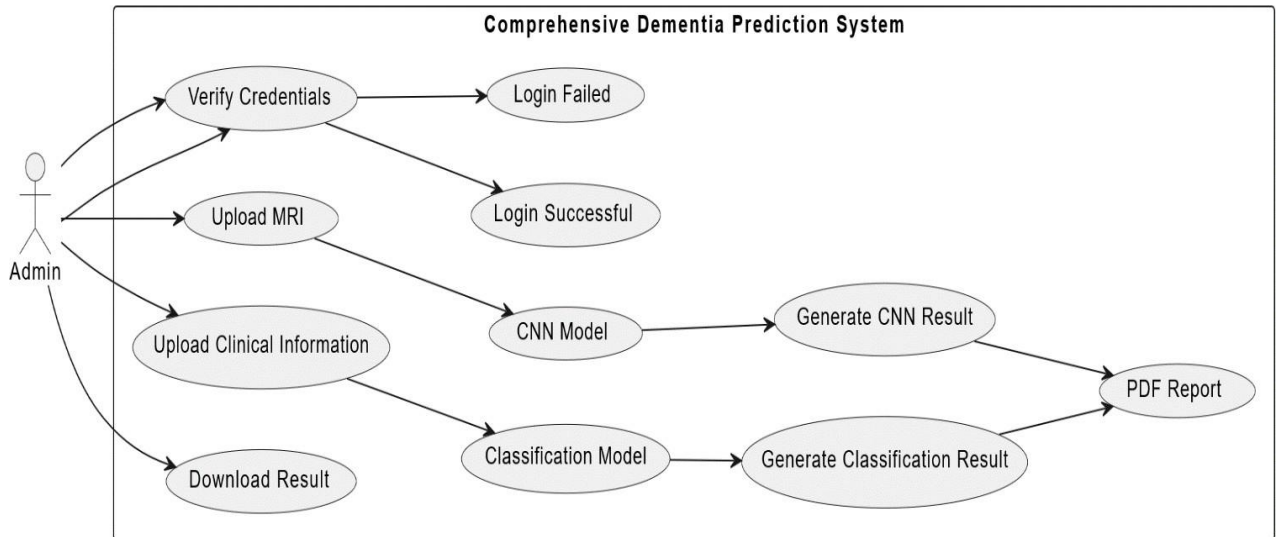


Fig 3.1: Use Case Diagram

The admin logs in and uploads MRI images and patient details. The system then carefully examines the data using separate models, combining their findings to provide a diagnosis. If there's any uncertainty, it suggests further tests. Finally, it creates a detailed report in PDF format, ready for the admin to share with healthcare professionals, ensuring collaborative decision-making.

3.3.2 Sequential Diagram

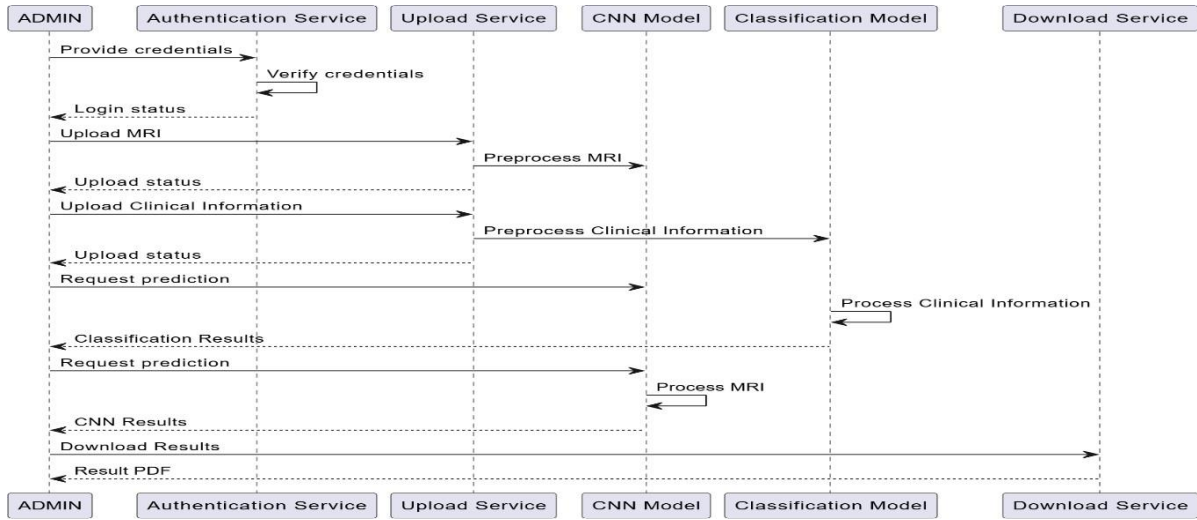


Fig 3.2: Sequential Diagram

First, the admin logs into the system using their unique credentials. Then, they upload MRI images and input patient information, like age and cognitive test scores. The system carefully examines this data using separate models, just like doctors analyzing different aspects of a patient's health. Once all the information is gathered, the system combines the findings to provide a comprehensive diagnosis. Finally, it generates a detailed report in PDF format, like a summary ready to be shared with healthcare professionals for further discussion and action.

3.3.3 Class Diagram

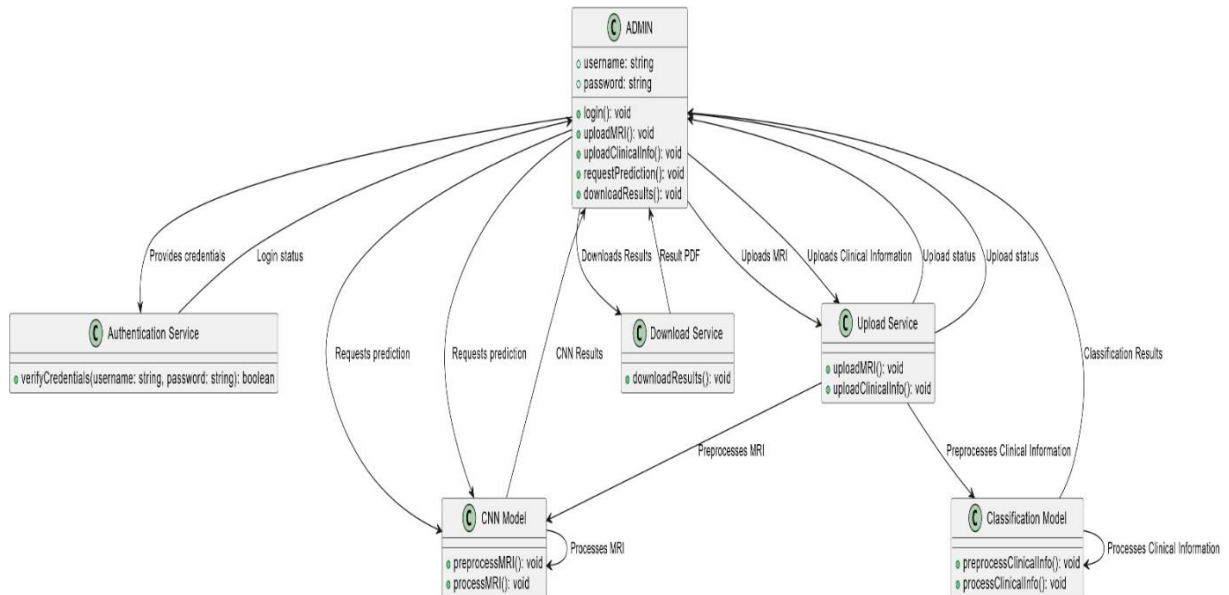


Fig 3.3: Class diagram

3.4 Methodology

3.4.1 Data Collection

a. MRI Dataset:

Acquire MRI image datasets from Kaggle categorized into multiple classes representing dementia severity (non-demented, very mild demented, mild demented, moderate demented).

b. Patient Attribute Dataset:

Collect patient attribute data including age, gender, MMSE, ASF, eTIV, CDR, normalized brain weight, and relevant attributes from reliable sources or clinical databases.

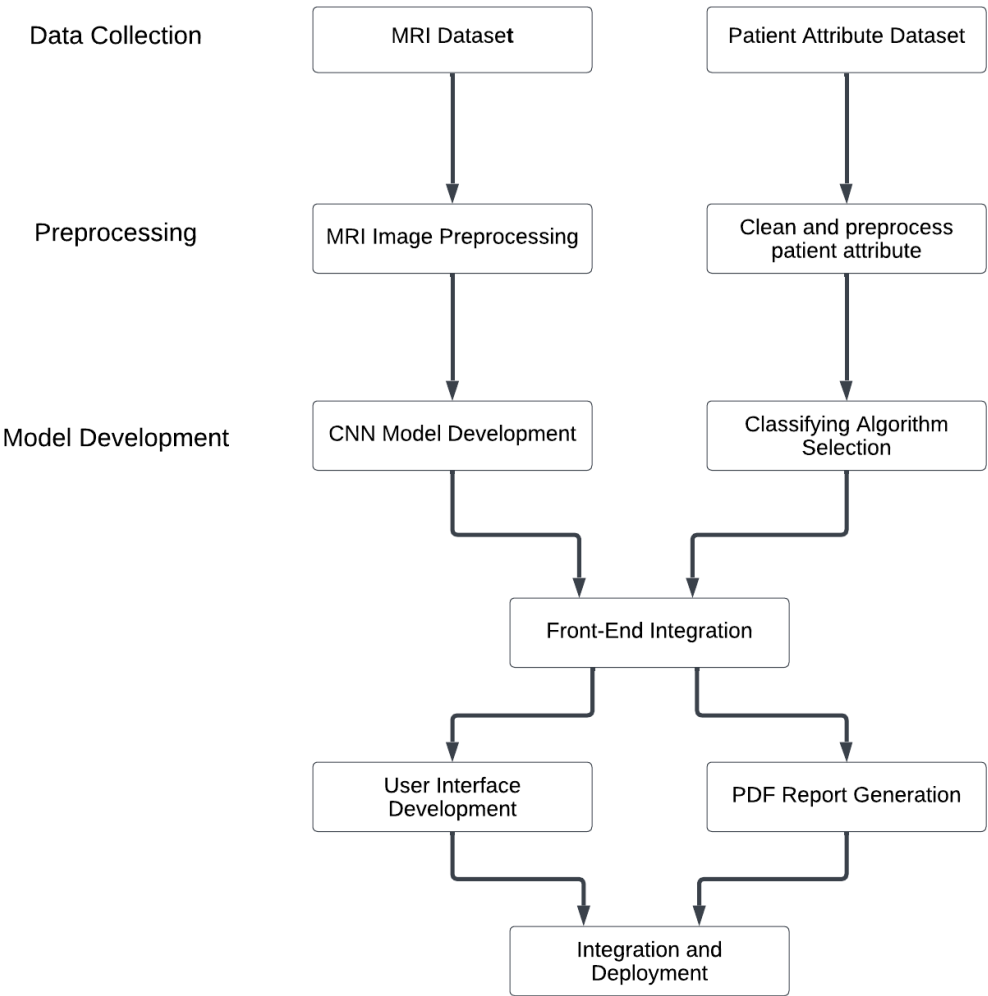


Fig: 3.4 Flow Diagram

3.4.2 Preprocessing

a. MRI Image Preprocessing:

Standardize image sizes and formats for consistency. Normalize pixel values, perform resizing, and apply techniques like augmentation to enhance the dataset for robust CNN model training.

b. Attribute dataset Preprocessing:

Clean and preprocess patient attribute data by handling missing values and normalizing numerical features for the classifying algorithm.

3.4.3 Model Development

For building our models, we take two approaches. First, for MRI images, we construct a deep learning model using TensorFlow and Keras. It's like crafting a puzzle with layers of filters and patterns to decode the images. We train it on diverse data, tweaking settings like brightness and angle to improve accuracy. Then, for patient attributes, we explore different algorithms, sort of like trying out various tools to solve a problem. We train and test each method on our cleaned-up data to find the best fit for predicting dementia.

3.4.4 User Interface Development

Design a user-friendly interface enabling the admin to upload MRI images for CNN analysis. Input patient details for the classifying algorithm model. Manage system functionalities and inputs seamlessly.

3.4.5 Report Generation

Develop a mechanism to compile comprehensive reports containing:

- Patient details.
- Results and findings from both the CNN and classifying algorithm models.
- Recommendations or further steps based on model outcomes.

Chapter 4

SYSTEM REQUIREMENTS AND SPECIFICATION

4.1 Hardware Requirements

- System: Intel I3.
- Hard Disk: 120 GB.
- Monitor: 15’’ LED
- Input Devices: Keyboard, Mouse
- RAM: 8 GB

4.2 Software Requirements

- Operating system: Windows 7.
- Coding Language: PYTHON
- Tools : Google Collab , VS Code

4.3 User Requirements

- Admin users who can upload MRI scans and input patient details into the system.
- Users with basic computer literacy to navigate the web application.
- Users interested in early detection and prediction of dementia.
- Users who understand the importance of timely interventions in healthcare.

4.4 System Requirements

- Integration of MRI-based predictive model using CNN.
- Integration of patient-recorded data predictive model using classification algorithms.
- Functional web application for admins to upload data and receive predictions.

- Validation mechanism to assess the accuracy and effectiveness of the predictive models.
- Secure data handling to protect patient information and maintain confidentiality.

4.5 Functional Requirements

- Develop and deploy separate prediction models based on MRI scans and patient data.
- Achieve a prediction accuracy of at least 85% for both models.
- Design an admin-friendly interface for uploading data and receiving predictions.
- Validate the effectiveness of integrating separate analyses for MRI and patient data.
- Ensure seamless data flow between the models within the integrated framework.

4.6 Non-Functional Requirements

- User should start the application.
- Performance: The system should provide timely predictions without significant delays.
- Security: Data confidentiality and integrity should be maintained throughout the process.
- Usability: The web application should be intuitive and easy to use for admins.
- Reliability: The system should consistently deliver accurate predictions under varying conditions.
- Scalability: The system should be able to handle an increasing amount of data and users over time.

Chapter 5

IMPLEMENTATION

5.1 General Implementation

We begin our implementation by laying the groundwork for the system. Next, we compile our data, much like assembling jigsaw pieces, to make sure we have all the information we need. We then preprocess the data, which is similar to tidying and arranging a disorganized space to make it easier to explore. Model development is what we do after our data is prepared. In this stage, we build our MRI image deep learning model and refine it like a master artisan polishing his masterpiece. In parallel, we investigate several patient attribute categorization algorithms, experimenting with different instruments to determine which one is most appropriate for the task. Following extensive training and testing, we assess each model's performance. Lastly, we adjust our models in response to input, making them better and more efficient over time. Our goal with this iterative process is to develop a solid and trustworthy method for thorough dementia detection.

5.2 Algorithms

5.2.1 Random Forest Classifier

Random Forest is an ensemble learning method that constructs a multitude of decision trees during training and outputs the mode of the classes (classification) or mean prediction (regression) of the individual trees. Each tree in the forest is trained on a random subset of the training data, and during prediction, the output of all trees is aggregated. This method reduces overfitting and increases robustness compared to individual decision trees.

5.2.2 Decision Tree Classifier

Decision Tree is a non-parametric supervised learning method used for classification and regression tasks. It works by recursively partitioning the feature space into smaller regions based on feature values, aiming to minimize impurity in each partition. This creates a tree-like structure where each leaf node represents a class label or a regression value. Decision trees are interpretable and can handle both numerical and categorical data.

5.2.3 XG Boost

XG Boost, short for Extreme Gradient Boosting, is an efficient implementation of gradient

boosting machines. It sequentially builds a series of decision trees, where each subsequent tree corrects the errors made by the previous one. XG Boost uses a regularized objective function to prevent overfitting and is known for its scalability and speed. It often outperforms other algorithms on structured/tabular data competitions.

5.2.4 Voting Classifier

A Voting Classifier combines multiple individual models trained on the same dataset and combines their predictions by majority voting (for classification) or averaging (for regression). It can be implemented with various algorithms such as decision trees, support vector machines, or logistic regression. Voting classifiers are particularly useful when individual models have different strengths and weaknesses, leading to improved overall performance.

5.2.5 Extra Tree Classifier

Extra Trees, short for Extremely Randomized Trees, is an ensemble learning method similar to Random Forest. However, it introduces additional randomness in the tree-building process by selecting random thresholds for each feature rather than searching for the best possible thresholds. This makes Extra Trees faster to train but may lead to slightly higher bias. It's a suitable choice when computational resources are limited.

5.2.6 Gradient Boosting Classifier

Gradient Boosting is a machine learning technique for regression and classification problems, which builds a model in a stage-wise fashion. It builds trees sequentially, where each tree corrects the errors of the previous one. Unlike other boosting algorithms, Gradient Boosting optimizes a differentiable loss function by adding weak learners in a greedy manner. This method has proven to be very effective and is widely used in various machine learning competitions.

5.2.7 AdaBoost Classifier

AdaBoost, short for Adaptive Boosting, is an ensemble learning method that combines multiple weak learners to create a strong classifier. It works by sequentially fitting models to repeatedly modified versions of the data. In each iteration, the algorithm assigns higher weights to incorrectly classified instances, forcing subsequent weak learners to focus more on difficult cases. AdaBoost is particularly effective in binary classification tasks and is less prone to overfitting.

5.2.8 LSTM (Long Short-Term Memory)

LSTM is a type of recurrent neural network (RNN) architecture designed to overcome the vanishing gradient problem in traditional RNNs. It introduces specialized memory cells that can

maintain information over long sequences, making it well-suited for tasks involving sequential data such as time series forecasting, natural language processing, and speech recognition. LSTM networks have a gating mechanism that allows them to selectively learn and forget information, enabling better long-term dependencies modeling.

5.2.9 BLSTM (Bidirectional Long Short-Term Memory)

BLSTM is an extension of LSTM that incorporates information from both past and future states by processing the input sequence in both forward and backward directions. This bidirectional processing allows the model to capture contextual information from both preceding and succeeding time steps, enhancing its ability to understand the temporal dependencies within the data. BLSTM networks are widely used in tasks such as speech recognition, machine translation, and sentiment analysis.

5.2.10 SVC

SVC is a supervised learning algorithm used for classification tasks. It works by finding the hyperplane that best separates the classes in the feature space, maximizing the margin between the classes. In cases where the data is not linearly separable, SVC uses a kernel trick to map the input space into a higher-dimensional feature space where separation is possible. SVC is effective in high-dimensional spaces and is versatile due to its ability to use different kernel functions.

5.2.11 Convolutional Neural Networks (CNN)

CNNs are a class of deep neural networks, most commonly applied to analyzing visual imagery. They are designed to automatically and adaptively learn spatial hierarchies of features from input images. CNNs consist of multiple layers, including convolutional layers, pooling layers, and fully connected layers. Convolutional layers apply filters to input images to extract local features, while pooling layers reduce spatial dimensions, helping to decrease computational complexity and overfitting. CNNs have shown remarkable success in various computer vision tasks, such as image classification, object detection, and semantic segmentation. Their ability to capture spatial hierarchies of features makes them suitable for tasks where understanding the spatial structure of data is crucial.

5.3 Packages Imported

1] Scikit-learn: Scikit-learn, also known as sklearn, is a Python library that provides simple and efficient tools for data mining and data analysis. It features various machine learning algorithms, including classification, regression, clustering, and dimensionality reduction, as well as tools for model selection and evaluation. Scikit-learn is designed to be user-friendly and accessible,

making it suitable for both beginners and experts in the field of machine learning.

2] Matplotlib: Matplotlib is a Python library for creating static, animated, and interactive visualizations in Python. It provides a comprehensive set of plotting tools for generating line plots, scatter plots, bar plots, histograms, and more. Matplotlib is widely used for data exploration, analysis, and presentation in scientific computing, data analysis, and machine learning projects.

3] NumPy: NumPy is a fundamental package for scientific computing in Python, providing support for multidimensional arrays, mathematical functions, linear algebra operations, and random number generation. It enables efficient manipulation of large datasets and facilitates numerical computations required for data analysis, machine learning, and scientific research.

4] Pandas: Pandas is a Python library for data manipulation and analysis, offering data structures and functions for efficiently handling structured data, such as tabular data and time series. It provides DataFrame and Series objects for data representation and offers tools for data selection, filtering, grouping, and merging, facilitating data exploration and preprocessing tasks.

5] ReportLab: ReportLab is a Python library for generating dynamic PDF documents programmatically. It provides tools for creating text, graphics, and tables in PDF format, allowing users to generate reports, invoices, and documents with precise control over layout, fonts, and content.

6] OpenCV: OpenCV(Open Source Computer Vision Library) is an open-source computer vision and machine learning software library. It provides wide range of functions and algorithms for image, video processing, including object detection, recognition, tracking, and segmentation. OpenCV is widely used in various applications, such as robotics, medical imaging etc.

5.4 Code Snippet

```
# Define the CNN model
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(image_height, image_width, 3)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(128, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(4, activation='softmax')
])
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])
model.fit(x_train, y_train, epochs=10, validation_data=(x_test, y_test))
```

Fig 5.1: Defining CNN Model

SVC

```
svm = SVC(kernel="linear", C=0.1, random_state=0)
svm.fit(X_train, y_train.ravel())
report_performance(svm)
roc_curves(svm)
accuracy(svm)
```

Fig 5.2: Defining SVM Model

```
from sklearn import metrics
def plot_confusion_matrix(y_test, model_test):
    cm = metrics.confusion_matrix(y_test, model_test)
    plt.figure(1)
    plt.clf()
    plt.imshow(cm, interpolation='nearest', cmap=plt.cm.Wistia)
    classNames = ['Nondemented', 'Demented']
    plt.title('Confusion Matrix')
    plt.ylabel('True label')
    plt.xlabel('Predicted label')
    tick_marks = np.arange(len(classNames))
    plt.xticks(tick_marks, classNames)
    plt.yticks(tick_marks, classNames)
    s = [['TN', 'FP'], ['FN', 'TP']]
    for i in range(2):
        for j in range(2):
            plt.text(j, i, str(s[i][j]) + " = " + str(cm[i][j]))
    plt.show()
```

Fig 5.3: Confusion Matrix

```

def report_performance(model):
    model_test = model.predict(X_test)

    print("\n\nConfusion Matrix:")
    print("{0}".format(metrics.confusion_matrix(y_test, model_test)))
    print("\n\nClassification Report: ")
    print(metrics.classification_report(y_test, model_test))
    #cm = metrics.confusion_matrix(y_test, model_test)
    plot_confusion_matrix(y_test, model_test)

total_fpr = {}
total_tpr = {}
def roc_curves(model):
    predictions_test = model.predict(X_test)
    fpr, tpr, thresholds = roc_curve(predictions_test, y_test)
    roc_auc = auc(fpr, tpr)
    total_fpr[str((str(model).split('(')[0])))] = fpr
    total_tpr[str((str(model).split('(')[0])))] = tpr
    plt.figure()
    plt.plot(fpr, tpr, color='darkorange', lw=1, label='ROC curve (area = %0.2f)' % roc_auc)
    plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.05])
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')
    plt.title('Receiver operating characteristic')
    plt.legend(loc="lower right")
    plt.show()

```

Fig 5.4: Performance report and ROC curve

```

total_accuracy = {}
def accuracy(model):
    pred = model.predict(X_test)
    accu = metrics.accuracy_score(y_test, pred)
    print("\nAccuracy Of the Model: ", accu, "\n\n")
    total_accuracy[str((str(model).split('(')[0])))] = accu

```

Fig 5.5: Accuracy

Chapter 6

RESULTS

6.1 Model Evaluation

Model evaluation is a critical step in the machine learning pipeline, as it allows for the assessment of a model's performance on unseen data. This process involves comparing predicted outcomes with ground truth labels using various evaluation metrics such as accuracy, precision, recall, F1 score, and ROC curve analysis

6.1.1 Performance Evaluation

MODEL	PRECISION	RECALL	F1 SCORE	SUPPORT	ACCURACY
Random Forest	0.95	0.95	0.95	75	0.946
Decision tree	0.95	0.95	0.95	75	0.946
Extra Tree	0.93	0.93	0.93	75	0.933
SVC	0.96	0.96	0.96	75	0.96
LSTM	-	-	-	-	0.93
XG Boost	0.95	0.95	0.95	75	0.946
AdaBoost	0.89	0.89	0.89	75	0.893
BLSTM	-	-	-	-	0.93
Voting classifier	0.95	0.95	0.95	75	0.946
Gradient Boosting	0.93	0.93	0.93	75	0.933

Table 6.1: Performance Metrics of different Models

6.1.2 Accuracy vs Model Comparison

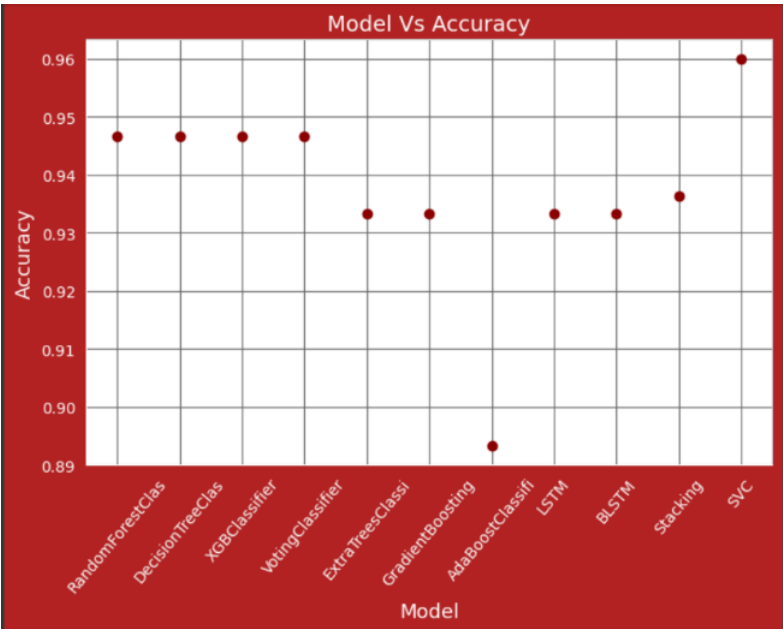


Fig 6.1: Accuracy vs Model graph

6.1.3 Confusion Matrix

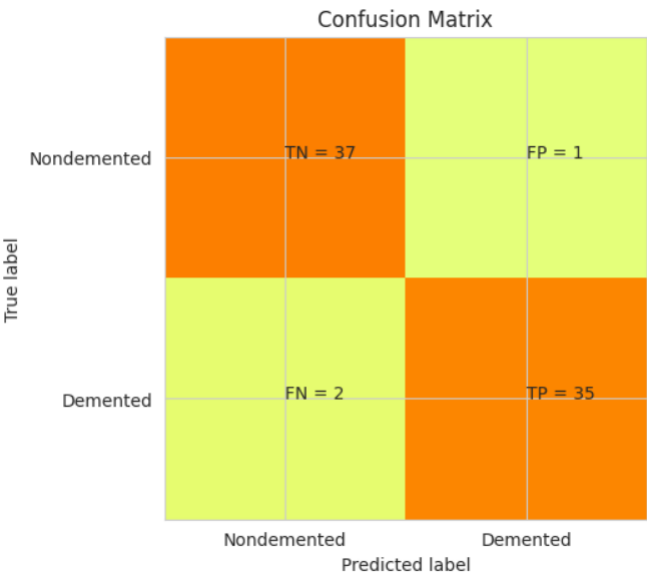


Fig 6.2: Confusion Matrix of SVM Model

6.1.4 ROC Curve and Accuracy

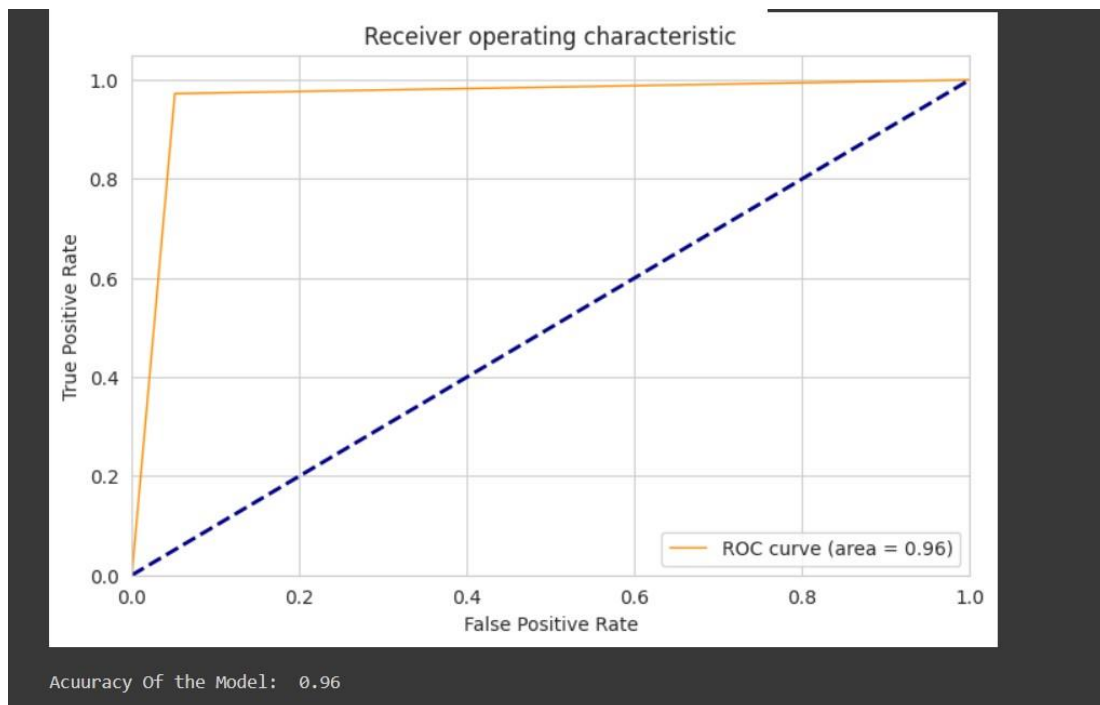


Fig 6.3: ROC Curve and Accuracy of SVM

6.1.5 Accuracy of CNN Model

```

Epoch 1/10
160/160 [=====] - 154s 952ms/step - loss: 0.9992 - accuracy: 0.5158 - val_loss: 0.8584 - val_accuracy: 0.6227
Epoch 2/10
160/160 [=====] - 146s 916ms/step - loss: 0.7977 - accuracy: 0.6453 - val_loss: 0.7693 - val_accuracy: 0.6273
Epoch 3/10
160/160 [=====] - 150s 937ms/step - loss: 0.5343 - accuracy: 0.7711 - val_loss: 0.4782 - val_accuracy: 0.8047
Epoch 4/10
160/160 [=====] - 149s 931ms/step - loss: 0.2574 - accuracy: 0.9006 - val_loss: 0.2917 - val_accuracy: 0.8844
Epoch 5/10
160/160 [=====] - 149s 930ms/step - loss: 0.1177 - accuracy: 0.9621 - val_loss: 0.2114 - val_accuracy: 0.9383
Epoch 6/10
160/160 [=====] - 150s 936ms/step - loss: 0.0464 - accuracy: 0.9846 - val_loss: 0.2614 - val_accuracy: 0.8992
Epoch 7/10
160/160 [=====] - 164s 1s/step - loss: 0.0136 - accuracy: 0.9967 - val_loss: 0.1060 - val_accuracy: 0.9695
Epoch 8/10
160/160 [=====] - 154s 966ms/step - loss: 0.0063 - accuracy: 0.9988 - val_loss: 0.0954 - val_accuracy: 0.9750
Epoch 9/10
160/160 [=====] - 156s 980ms/step - loss: 0.0594 - accuracy: 0.9789 - val_loss: 0.3095 - val_accuracy: 0.9016
Epoch 10/10
160/160 [=====] - 167s 1s/step - loss: 0.0253 - accuracy: 0.9920 - val_loss: 0.1082 - val_accuracy: 0.9766
<keras.src.callbacks.History at 0x7ce94e4529e0>

```

Fig 6.4: Accuracy of CNN Model

6.2 User Interface

Our user interface simplifies the process of interacting with our system. It starts with a familiar login page, providing a secure entry point for authorized users. From there, users are seamlessly guided to the data input page, where they can easily upload MRI images and input patient details, much like filling out a standard form. Once the data is submitted, users are directed to the result page, where they can view the analysis outcome presented in a clear and understandable format. This interface design prioritizes user experience, ensuring accessibility and ease of use throughout the entire process.

6.2.1 Login Page

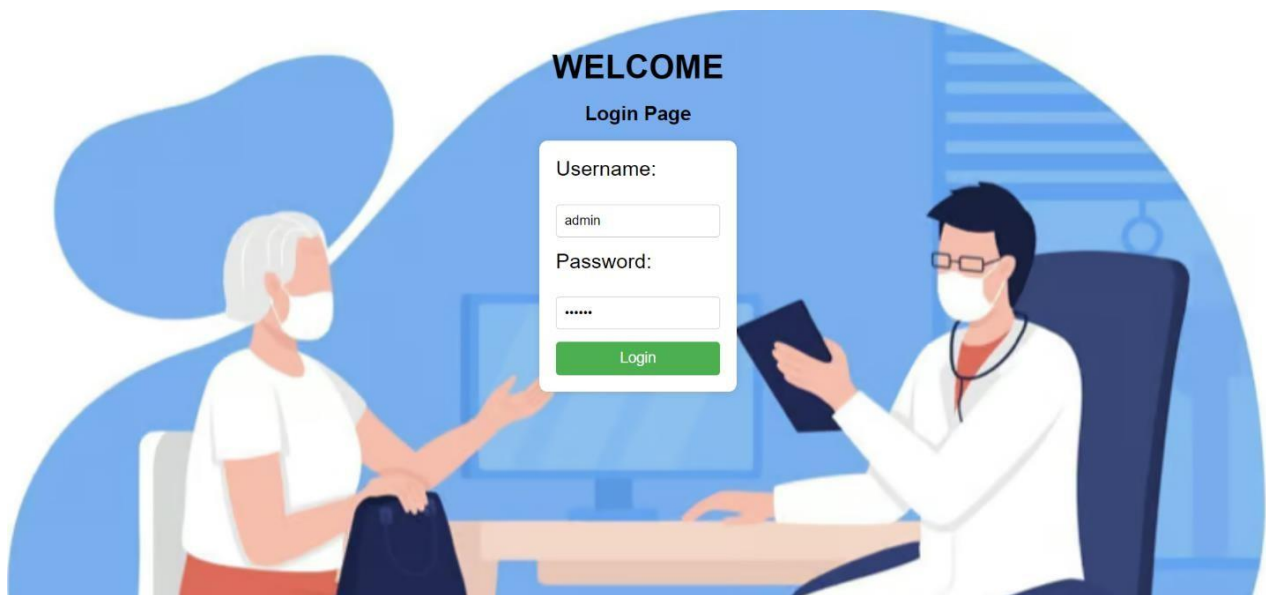
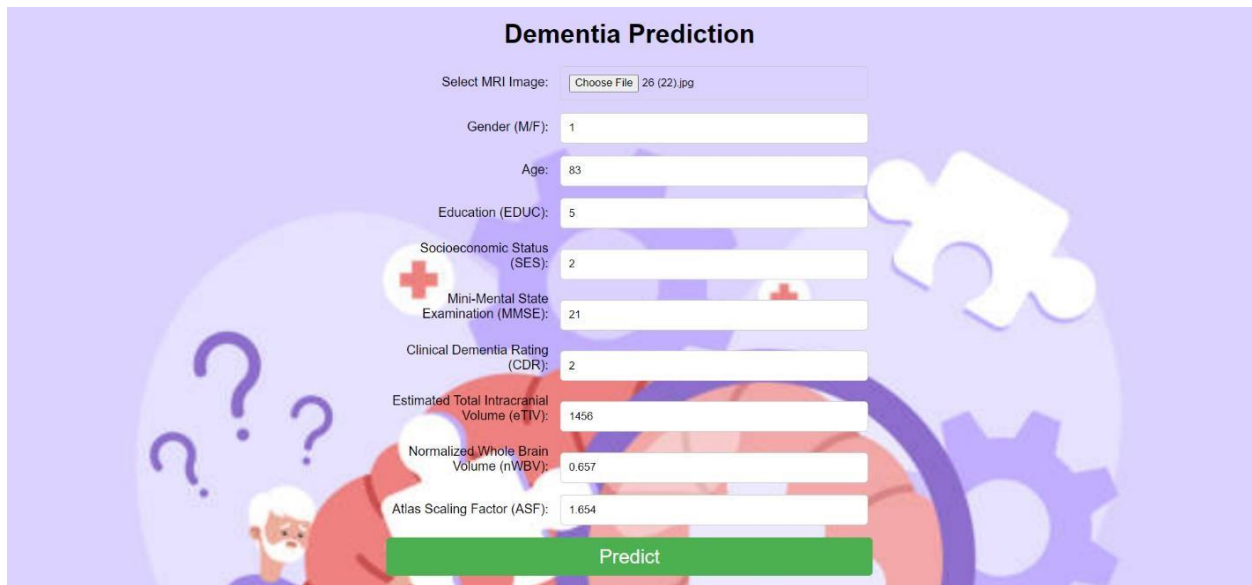


Fig 6.5: Login page

Login page asks for username and password. This will be known only to Admin there by maintaining the Confidentiality and Security of data and model. If the Admin enters the wrong password he wont be logged in . Once right username and password are entered he will be moved to Input page.

6.2.2 Input Page



The image shows a web form titled "Dementia Prediction" on a light purple background with a brain and puzzle pieces illustration. The form contains several input fields for patient data and a "Predict" button.

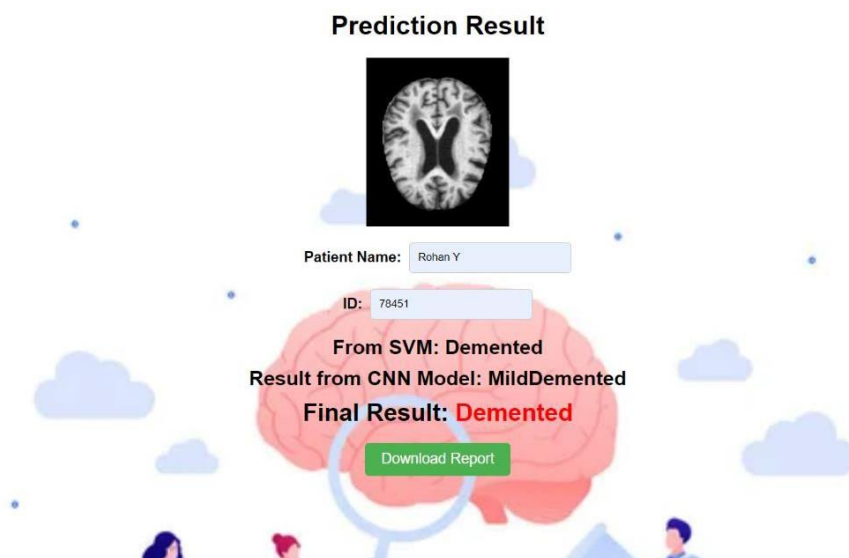
Field	Value
Select MRI Image:	Choose File 26 (22).jpg
Gender (M/F):	1
Age:	83
Education (EDUC):	5
Socioeconomic Status (SES):	2
Mini-Mental State Examination (MMSE):	21
Clinical Dementia Rating (CDR):	2
Estimated Total Intracranial Volume (eTIV):	1456
Normalized Whole Brain Volume (nWBV):	0.657
Atlas Scaling Factor (ASF):	1.654

Predict

Fig 6.6: Input Page

Once successfully logged in , admin will be provided with an input page , where admin must select the patient's digital MRI scan to be uploaded and enter the patient details. When all the information is entered admin can just click on the predict button to see the results.

6.2.3 Result Page



The image shows a web page titled "Prediction Result" with a brain scan image and patient information. The results from two models are displayed, along with a combined "Final Result" and a "Download Report" button.

Field	Value
Patient Name:	Rohan Y
ID:	78451
From SVM:	Demented
Result from CNN Model:	MildDemented
Final Result:	Demented

Download Report

Fig 6.7: Result Page

Result page provides the result of both models along with the combined result. Amdmin will be able to print the report and later provide it to patients.

Chapter 7

CONCLUSION AND FUTURE ENHANCEMENT

The Comprehensive Dementia Detection system developed is a significant advancement in the field of healthcare technology. By integrating MRI-based predictive models using Convolutional Neural Networks (CNN) and patient-recorded data predictive models using classification algorithms, the system offers a reliable and efficient way to detect and predict dementia at an early stage.

The user requirements, system requirements, and functional requirements outlined in the report emphasize the importance of user-friendly interfaces for administrators, accurate prediction models, and secure data handling to protect patient confidentiality. The system's design ensures a seamless flow of data, accurate predictions, and timely interventions, reflecting a thoughtful consideration for both accuracy and patient privacy.

The validation mechanism implemented to assess the accuracy and effectiveness of the predictive models, along with the emphasis on data security and confidentiality, further enhance the system's credibility and reliability. The non-functional requirements such as performance, security, usability, reliability, and scalability ensure that the system can deliver accurate predictions under varying conditions while maintaining data integrity and confidentiality.

Further Enhancements

1] Integration of Advanced Technologies: Explore the integration of emerging technologies such as Natural Language Processing (NLP) and Reinforcement Learning to enhance the system's capabilities in analyzing patient data and providing personalized recommendations.

2] Mobile Application Development: Develop a mobile application version of the system to enable easy access for users and caregivers, allowing them to input data, receive predictions, and track progress conveniently on their smartphones.

3] Longitudinal Data Analysis: Implement longitudinal data analysis techniques to track disease progression over time, enabling early detection of cognitive decline and facilitating personalized treatment plans for better patient outcomes.

Chapter 8

BIBLIOGRAPHY

- [1] Jain, Varun, et al, A Novel DCGAN-based Augmentation and Classification (D-BAC) Model Approach for Detection and Severity Prediction of Dementia Using MRI, *NeuroImage*, vol. 9, Nov. 2021, pp. 1-23.
- [2] Ho SY-C, Chien T-W, Lin M-L, Tsai K-T, An app for predicting patient dementia classes using convolutional neural networks (CNN) and artificial neural networks (ANN): Comparison of prediction accuracy in Microsoft Excel, *Medicine* .2023 ;102:4(e32670).
- [3] Dhakal, S., Azam, S., & Khan, M. (), Dementia Prediction Using Machine Learning. *Procedia Computer Science*, 2023, 219, 1297-1308.
- [4] Sharma, Deep et al. Predem: A computational framework for prediction of early dementia using deep neural networks, *Procedia Computer Science*, Vol 215, 2022, pp. 697-705.
- [5] Ali, L., Khan, S.U., Arshad, M., Ali, S., Anwar, M., Machine Learning for Dementia Prediction: A Systematic Review, *Journal of Medical Systems*, Vol. 47, Issue 17, 2023, pp. 1-25.
- [6] Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, *Introduction to Statistical Learning*, Springer, New York, NY, USA, 1st Edition, 2013.
- [7] Andreas C. Müller and Sarah Guido, *Introduction to Machine Learning with Python*, O'Reilly Media, Sebastopol, CA, USA, 1st Edition, 2016
- [8] Tianqi Chen and Carlos Guestrin, *XGBoost: A Scalable Tree Boosting System*, ACM Press, New York, NY, USA, 1st Edition, 2016
- [9] Sebastian Raschka and Vahid Mirjalili, *Python Machine Learning*, Packt Publishing, Birmingham, UK, 2019.
- [10] Fabian Pedregosa et al., *Scikit-Learn: Machine Learning in Python*, Springer, New York, NY, USA, 1st Edition, 2011.
- [11] Trevor Hastie, Robert Tibshirani, and Jerome Friedman, *Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Springer, New York, NY, USA, 2nd Edition, 2009
- [12] Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, New York, NY, USA, 1st Edition, 2006
- [13] Ian Goodfellow, Yoshua Bengio, and Aaron Courville, *Deep Learning*, MIT Press,

Cambridge, MA, USA, 1st Edition, 2016.

[14] Palash Goyal et al., Deep Learning for Natural Language Processing, O'Reilly Media, Sebastopol, CA, USA, 1st Edition, 2018.

[15] Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, Sebastopol, CA, USA, 2nd Edition, 2019.

[16] Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, Sebastopol, CA, USA, 1st Edition, 2016.

[17] Wes McKinney, Python for Data Analysis, O'Reilly Media, Sebastopol, CA, USA, 2nd Edition, 2017.

[18] Dusty Phillips, Python 3 Object-Oriented Programming, Packt Publishing, Birmingham, UK, 2nd Edition, 2015.

[19] Joseph Howse, Joe Minichino, and Prateek Joshi, Learning OpenCV 4 with Python, Packt Publishing, Birmingham, UK, 1st Edition, 2018.



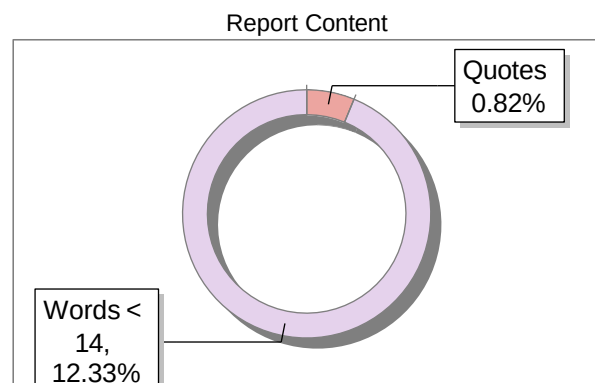
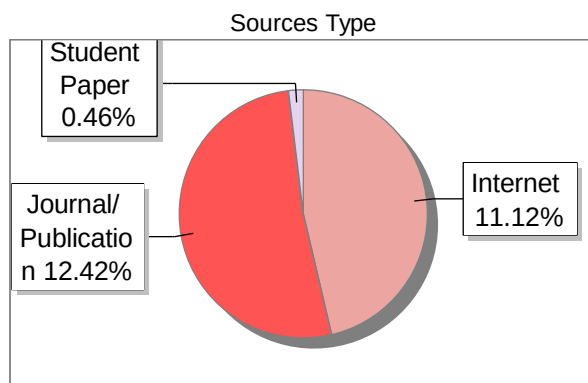
The Report is Generated by DrillBit Plagiarism Detection Software

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Title	Comprehensive dementia detection:MRI Insights and Patient Information
Paper/Submission ID	1793808
Submitted by	*****
Submission Date	*****
Total Pages, Total Words	29, 6733
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19	springeropen.com	<1	Internet Data
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23	Multi-split optimized bagging ensemble model selection for multi-class education by Injadat-2020	<1	Publication
24	nexocode.com	<1	Internet Data
25	springeropen.com	<1	Internet Data
26	www.fcc.gov	<1	Publication
27	cssjournal.org	<1	Internet Data
28	mdpi.com	<1	Internet Data
29	www.dx.doi.org	<1	Publication
30	bohrpub.com	<1	Internet Data
31	Submitted to Visvesvaraya Technological University, Belagavi	<1	Student Paper
32	www.linkedin.com	<1	Internet Data

33	www.mdpi.com	<1	Internet Data
34	www.ncbi.nlm.nih.gov	<1	Internet Data
35	FIELD EVALUATION OF AN ONLINE FOSTER PARENT TRAINING SYSTEM by BUZHARDT-2006	<1	Publication
36	www.frontiersin.org	<1	Internet Data
37	d-nb.info	<1	Publication
38	docplayer.net	<1	Internet Data
39	iisuniv.ac.in	<1	Publication
40	journals.plos.org	<1	Publication
41	Neural Network Based Brain Tumor Detection Using Wireless Infrared Im, by Shakeel, P. Mohamed- 2019	<1	Publication
42	philpapers.org	<1	Internet Data
43	springeropen.com	<1	Internet Data
44	springeropen.com	<1	Internet Data
45	www.frontiersin.org	<1	Internet Data
46	www.linkedin.com	<1	Internet Data
47	www.ncbi.nlm.nih.gov	<1	Internet Data
48	archivesofmedicalsecience.com	<1	Publication
49	Ceilometer inversion method using water-vapor correction from co-located microwa by Bedoya-Velsquez-2021	<1	Publication
50	www.cbit.ac.in	<1	Publication

51	An extension of the FURIA classification algorithm to low quality data through f by Palacios-2016	<1	Publication
52	azpdf.tips	<1	Internet Data
53	dovepress.com	<1	Internet Data
54	dspace.lib.cranfield.ac.uk	<1	Publication
55	famt.ac.in	<1	Publication
56	Machine Learning in Rheumatic Diseases by Jiang-2020	<1	Publication
57	moam.info	<1	Internet Data
58	pdfs.semanticscholar.org	<1	Publication
59	sportdocbox.com	<1	Internet Data
60	springeropen.com	<1	Internet Data
61	www.ieindia.org	<1	Publication
62	www.rationalrevolution.net	<1	Internet Data
63	www.yumpu.com	<1	Internet Data
64	1library.co	<1	Internet Data
65	artsdocbox.com	<1	Internet Data
66	asbmr.onlinelibrary.wiley.com	<1	Internet Data
67	Bioremediation of Cephalexin with non-living Chlorella sp., biomass, by Angulo, E.; Bula, L - 2018	<1	Publication
68	Cloud-assisted Industrial Internet of Things (IIoT)- enabled framework by Hossain-2016	<1	Publication

69	digitalcommons.uri.edu	<1	Publication
70	docplayer.net	<1	Internet Data
71	etd.aau.edu.et	<1	Publication
72	Graph theory and network topological metrics may be the potential biomarker in P by Huang-2019	<1	Publication
73	inba.info	<1	Internet Data
74	journal.unair.ac.id	<1	Internet Data
75	journals.pan.pl	<1	Internet Data
76	link.springer.com	<1	Internet Data
77	Matrix-assisted laser desorption ionizationtime-of-flight mass spect, by Yi-Tzu Cho Hung Su- 2013	<1	Publication
78	medium.com	<1	Internet Data
79	Natural language inquiry to an open-ended data library by Potts-1970	<1	Publication
80	nature.com	<1	Internet Data
81	openlibrarypublications.telkomuniversity.ac.id	<1	Publication
82	Overview of Anomaly Detection techniques in Machine Learning by Toshniwal-2020	<1	Publication
83	philpapers.org	<1	Internet Data
84	Thesis submitted to dspace.mit.edu	<1	Publication
85	Vertical inequity in the tax assessment of lodging properties by Marcu-2010	<1	Publication

86	Voices to reckon with perceptions of voice identity in clinical and non-clinical by Badcock-2013	<1	Publication
87	www.cosis.net	<1	Publication
88	www.dx.doi.org	<1	Publication
89	www.dx.doi.org	<1	Publication
90	www.dx.doi.org	<1	Publication
91	www.geeksforgeeks.org	<1	Internet Data
92	www.linkedin.com	<1	Internet Data
93	www.mdpi.com	<1	Publication
94	ACM Press the 8th International Conference- Islamabad, Pakistan (2, by ur Rahman, Syed Saif- 2010	<1	Publication